Moving Objects Have Energy

Chapter 2 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **How are energy, change, and movement of objects related?**

NOTICE

In our marble demo, write what happened when the moving marble reached the target.

KEY WORDS

motion — Motion is the process of an object changing _____ .

energy of motion — Energy of motion is the energy an object has _____ it is moving.

MODEL IT

A moving object can cause another object to change. The change is our evidence.

MOVING OBJECT → **CAUSES CHANGE** → **EVIDENCE OF ENERGY**

DRAW / LABEL

Draw the marble before and after it hits the target. Use arrows to show the direction of motion.

BEFORE	AFTER
marble rolls toward target	*target moves after contact*

What changed? _____

Evidence + Lab Connection

Chapter 2 Guided Notes · Page 2

Name: _____ **Date:** _____

OBSERVATION VS INFERENCE

Observation = what I saw. Inference = what I think it means.

OBSERVATION (WHAT I SAW)	INFERENCE (WHAT I THINK)
The target marble moved.	The moving marble had energy.

EXAMPLE / NON-EXAMPLE

SHOWS ENERGY OF MOTION (EXAMPLE)	DOES NOT (NON-EXAMPLE)

LAB CONNECTION — MOVING MARBLES

Record one thing you observed and the evidence it gives.

WHAT I OBSERVED	EVIDENCE THAT THE MOVING OBJECT HAD ENERGY

CHECK YOUR THINKING

How do we know a moving object has energy, even though we cannot see the energy?

CONNECT *Chapter 1 → 2: energy can cause change; a moving object is one thing that has energy.*